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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (CST) (CBCS) Degree Examinations
 January 2023

22CST102 - STRUCTURAL DYNAMICS

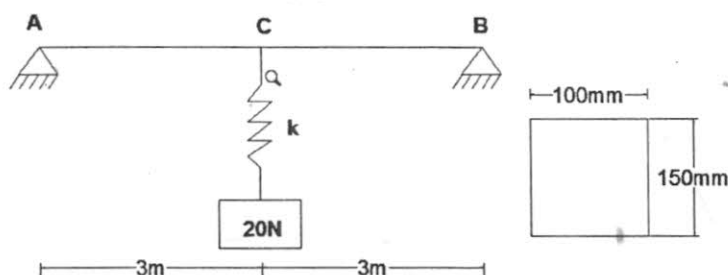
Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|--------------|------------|------------|------------|
| 1. a) Differentiate between Static and Dynamic Load with examples. | 10 | L*2 | 1 | 1 |
| b) Define various types of dynamic load with neat diagrams. | 10 | L2 | 1 | 1 |
| 2. a) What is degree of freedom? And how to express degree of freedom in dynamics. | 10 | L2 | 2 | 2 |
| b) Find the natural frequency of the system shown in fig assuming the beam mass as negligible, take $E=2.1 \times 10^5 \text{ N/mm}^2$. | | | | |



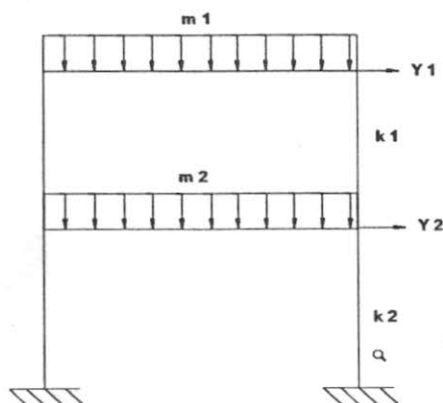
10 L3 2 3

Unit – II

- | | | | | |
|--|----|----|---|---|
| 3. Derive equation of motion for damped free vibration of SDOF system and also derive equation of motion for critically damped, over damped and under damped system. | 20 | L2 | 2 | 3 |
| 4. Derive response of damped system under base excitation. | 20 | L2 | 2 | 3 |

Unit – III

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|---|----|----|---|---|
| 5. a) Write a mathematical model for MDOF system. | 10 | L3 | 3 | 3 |
| b) Determine natural frequency of shear building shown, where $m_1=11564.6 \text{ kg}$, $m_2=23809.5 \text{ kg}$, $k_1=7.75 \times 10^6 \text{ N/m}$, $k_2=5.37 \times 10^6 \text{ N/m}$. | | | | |

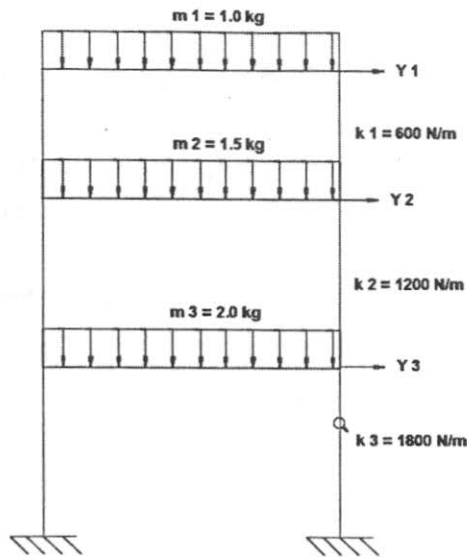


10 L3 3 3

P.T.O.

6. a) What is Mode shapes Explain? And discuss various characterization of mode.
b) Determine the natural frequency and mode shape for the frame shown.

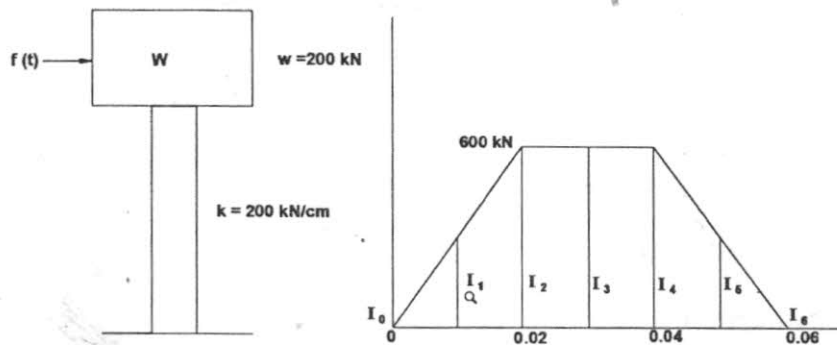
10 L3 3 2



10 L3 3 3

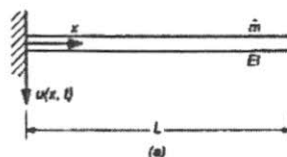
Unit - IV

7. Determine response of tower subjected to blast loading using Duhamel's Integral method.



20 L3 4 3

8. Find the natural frequency of vibration of the cantilever beam shown in figure using shape function $\psi_0(x) = (x/L)^2$ obtain more accurate estimate of the frequency by Improved Rayleigh's Method.



20 L3 4 3

Unit - V

9. What are flexural vibrations? Write natural vibration and mode shapes for beams with different end conditions.
10. Discuss any one laboratory experiment in dynamics.

20 L2 4 3

20 L3 5 4